

Q1 JEE Main 2020 - 7 Jan (Morning)

If $f(a + b + 1 - x) = f(x)$, for all x , where a and b are fixed positive real numbers, then

$\frac{1}{(a+b)} \int_a^b x (f(x) + f(x+1)) dx$ is equal to

(A) $\int_{a+1}^{b+1} f(x) dx$

(B) $\int_{a+1}^{b+1} f(x+1) dx$

(C) $\int_{a-1}^{b-1} f(x) dx$

(D) $\int_{a-1}^{b-1} f(x+1) dx$

Q2 JEE Main 2020 - 7 Jan (Evening)

The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is :

(A) $\ln 2$

(B) $\ln \sqrt{2}$

(C) $\ln \frac{3}{4}$

(D) $\ln \frac{4}{3}$

Q3 JEE Main 2020 - 7 Jan (Evening)

If θ_1 and θ_2 be respectively the smallest and the largest values of θ in $(0, 2\pi) - (\pi)$ which satisfy

the equation, $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$, then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is equal to :

(A) $\frac{\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{\pi}{9}$

(D) $\frac{\pi}{3} + \frac{1}{6}$

Q4 JEE Main 2020 - 8 Jan (Evening)

Let $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$ then

- (A) $\frac{1}{9} < I^2 < \frac{1}{8}$
- (B) $\frac{1}{3} < I^2 < \frac{1}{2}$
- (C) $\frac{1}{9} < I < \frac{1}{8}$
- (D) $\frac{1}{3} < I < \frac{1}{2}$

Q5 JEE Main 2020 - 9 Jan (Morning)

Find the value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$

- (A) π^2
- (B) $2\pi^2$
- (C) $3\pi^2$
- (D) $4\pi^2$

Q6 JEE Main 2020 - 9 Jan (Morning)

If $f(x) = a + bx + cx^2$ where $a, b, c \in R$ then $\int_0^1 f(x) dx$ is

- (A) $\frac{1}{3} (f(1) + f(0) + 2f(\frac{1}{2}))$
- (B) $\frac{1}{6} (f(1) + f(0) + 4f(\frac{1}{2}))$
- (C) $\frac{1}{6} (f(1) + f(0) - 4f(\frac{1}{2}))$
- (D) $\frac{1}{6} (f(1) - f(0) - 4f(\frac{1}{2}))$

Q7 JEE Main 2020 - 9 Jan (Morning)

If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $f(0) = 0$ then the value of $f(1)$ is

- (A) $\frac{\pi+1}{4}$
- (B) $\frac{\pi-1}{4}$
- (C) $\frac{\pi+1}{2}$
- (D) 0

Q8 JEE Main 2020 - 2 Sep (Morning)

The integral $\int_0^2 ||x-1|-x| dx$ is equal to

Q9 JEE Main 2020 - 2 Sep (Evening)

Let $[t]$ denote the greatest integer less than or equal to t . Then the value of $\int_1^2 |2x - [3x]| dx$ is

Q10 JEE Main 2020 - 3 Sep (Morning)

$\int_{-\pi}^{\pi} |\pi - |x|| dx$ is equal to :

- (A) π^2
- (B) $2\pi^2$
- (C) $\sqrt{2}\pi^2$
- (D) $\frac{\pi^2}{2}$

Q11 JEE Main 2020 - 3 Sep (Evening)

If the value of the integral $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ is $\frac{k}{6}$, then k is equal to :

- (A) $2\sqrt{3} + \pi$
- (B) $3\sqrt{2} - \pi$
- (C) $3\sqrt{2} + \pi$
- (D) $2\sqrt{3} - \pi$

Q12 JEE Main 2020 - 4 Sep (Morning)

Let $f(x) = |x - 2|$ and $g(x) = f(f(x))$, $x \in [0, 4]$ Then $\int_0^3 (g(x) - f(x)) dx$ is equal to :

- (A) 0
- (B) $\frac{3}{2}$
- (C) $\frac{1}{2}$
- (D) 1

Q13 JEE Main 2020 - 4 Sep (Evening)

The integral $\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ is equal to:

- (A) $\frac{7}{18}$
- (B) $\frac{9}{2}$
- (C) $-\frac{1}{18}$
- (D) $-\frac{1}{9}$

Q14 JEE Main 2020 - 4 Sep (Evening)

Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\leq x$ respectively of a real number x . If $\int_0^n \{x\} dx$, $\int_0^n [x] dx$ and $10(n^2 - n)$, $(n \in N, n > 1)$ are three consecutive terms of a G.P., then n is equal to

Q15 JEE Main 2020 - 5 Sep (Morning)

The value of $\int_{-\pi/2}^{\pi/2} \frac{1}{1+e^{\sin x}} dx$ is :

- (A) $\frac{\pi}{4}$
- (B) π
- (C) $\frac{3\pi}{2}$
- (D) $\frac{\pi}{2}$

Q16 JEE Main 2020 - 6 Sep (Morning)

If $I_1 = \int_0^1 (1 - x^{50})^{100} dx$ and $I_2 = \int_0^1 (1 - x^{50})^{101} dx$ such that $I_2 = \alpha I_1$ then α equals to :

- (A) $\frac{5051}{5050}$
- (B) $\frac{5050}{5051}$
- (C) $\frac{5050}{5049}$
- (D) $\frac{5049}{5050}$

Q17 JEE Main 2020 - 6 Sep (Evening)

The integral $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ equals:

- (A) $e(2e - 1)$
- (B) $e(4e - 1)$
- (C) $4e^2 - 1$
- (D) $e(4e + 1)$

Answer Key

Q1 (D)	Q2 (A)	Q3 (A)	Q4 (A)
Q5 (A)	Q6 (B)	Q7 (A)	Q8 (1.5)
Q9 (1)	Q10 (A)	Q11 (D)	Q12 (D)
Q13 (C)	Q14 (21)	Q15 (D)	Q16 (B)
Q17 (B)			

Q1

$$I = \frac{1}{(a+b)} \int_a^b x [f(x) + f(x+1)] dx \quad \dots (1)$$

$$I = \frac{1}{(a+b)} \int_a^b (a+b-x) [f(a+b-x) + f(a+b+1-x)] dx$$

$$I = \frac{1}{(a+b)} \int_a^b (a+b-x) [f(x+1) + f(x)] dx \quad \dots (2)$$

[$\because$ put $x \rightarrow x+1$ in given equation]

(1) + (2)

$$2I = \int_a^b [f(x+1) + f(x)] dx$$

$$2I = \int_a^b f(x+1) dx + \int_a^b f(x) dx$$

$$\int_a^b f(a+b+1-x) dx + \int_a^b f(x) dx$$

$$2I = 2 \int_a^b f(x) dx$$

$$I = \int_a^b f(x) dx$$

put $x = t+1 \Rightarrow I = \int_{a-1}^{b-1} f(t+1) dt$

Q2

$$4\alpha \left\{ \int_{-1}^0 e^{\alpha x} dx + \int_0^2 e^{-\alpha x} dx \right\} = 5$$

$$\Rightarrow 4\alpha \left\{ \left(\frac{e^{\alpha x}}{\alpha} \right)_{-1}^0 + \left(\frac{e^{-\alpha x}}{-\alpha} \right)_{0}^2 \right\} = 5$$

$$\Rightarrow 4\alpha \left\{ \left(\frac{1 - e^{-\alpha}}{\alpha} \right) - \left(\frac{e^{-2\alpha} - 1}{\alpha} \right) \right\} = 5$$

$$\Rightarrow 4(2 - e^{-\alpha} - e^{-2\alpha}) = 5 \quad \text{Put } e^{-\alpha} = t$$

$$\Rightarrow 4t^2 + 4t - 3 = 0$$

$$\Rightarrow (2t + 3)(2t - 1) = 0$$

$$\Rightarrow e^{-\alpha} = \frac{1}{2}$$

$$\Rightarrow \alpha = \ln 2$$

Q3

$$2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$$

$$\frac{2\cos^2\theta}{\sin^2\theta} - \frac{5}{\sin\theta} + 4 = 0$$

$$2\cos^2\theta - 5\sin\theta + 4\sin^2\theta = 0, \quad \sin\theta \neq 0$$

$$2\sin^2\theta - 5\sin\theta + 2 = 0$$

$$(2\sin\theta - 1)(\sin\theta - 2) = 0$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \int_{\pi/6}^{5\pi/6} \cos^2 3\theta d\theta = \int_{\pi/6}^{5\pi/6} \frac{1 + \cos 6\theta}{2} d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin 6\theta}{6} \right]_{\pi/6}^{5\pi/6} = \frac{1}{2} \left[\frac{5\pi}{6} - \frac{\pi}{6} + \frac{1}{6}(0 - 0) \right]$$

$$= \frac{1}{2} \cdot \frac{4\pi}{6} = \frac{\pi}{3}$$

Q4

$$f(x) = \frac{1}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$$

$$f'(x) = \frac{-1}{2} \frac{(6x^2 - 18x + 12)}{(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$= \frac{-6(x-1)(x-2)}{2(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$f(1) = \frac{1}{3}, \quad f(2) = \frac{1}{\sqrt{8}}$$

$$\frac{1}{3} < I < \frac{1}{\sqrt{8}}$$

Q5

$$\int_0^{\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} + \frac{(2\pi - x) \sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= \int_0^{\pi} \frac{2\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= 2\pi \int_0^{\pi/2} \frac{\sin^8 x}{\sin^8 x + \cos^8 x} + \frac{\cos^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= 2\pi \int_0^{\pi/2} 1 dx = 2\pi \times \frac{\pi}{2} = \pi^2$$

Q6

$$\int_0^1 (a + bx + cx^2) dx = ax + \frac{bx^2}{2} + \frac{cx^3}{3} \Big|_0^1 = a + \frac{b}{2} + \frac{c}{3}$$

$$f(1) = a + b + c$$

$$f(0) = a$$

$$f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$$

$$\text{Now } \frac{1}{6} \left(f(1) + f(0) + 4f\left(\frac{1}{2}\right) \right)$$

$$= \frac{1}{6} \left(a + b + c + a + 4 \left(a + \frac{b}{2} + \frac{c}{4} \right) \right)$$

$$= \frac{1}{6} (6a + 3b + 2c) = a + \frac{b}{2} + \frac{c}{3}$$

Q7

$$f'(x) = \tan^{-1}(\sec x + \tan x) = \tan^{-1}\left(\frac{1 + \sin x}{\cos x}\right)$$

$$= \tan^{-1}\left(\frac{1 - \cos\left(\frac{\pi}{2} + x\right)}{\sin\left(\frac{\pi}{2} + x\right)}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{4} + \frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4} + \frac{x}{2}\right)\cos\left(\frac{\pi}{4} + \frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right) = \frac{\pi}{4} + \frac{x}{2}$$

$$(f'(x)) dx = \frac{\pi}{4} + \frac{x}{2} dx$$

$$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + c$$

$$f(0) = c = 0 \Rightarrow f(x) = \frac{\pi}{4}x + \frac{x^2}{4}$$

$$\text{So } f(1) = \frac{\pi + 1}{4}$$

Q8

$$I = \int_0^1 |1 - x - x| dx + \int_1^2 |x - 1 - x| dx$$

$$= \int_0^1 |1 - 2x| dx + \int_1^2 dx$$

$$\begin{aligned}
 &= \int_0^{\frac{1}{2}} (1 - 2x) dx + \int_{\frac{1}{2}}^1 (2x - 1) dx + \int_1^2 dx \\
 &= \left[x - x^2 \right]_0^{\frac{1}{2}} + \left[x^2 - x \right]_{\frac{1}{2}}^1 + (2 - 1) \\
 &= \left(\frac{1}{2} - \frac{1}{4} \right) + (-) \left[\frac{1}{4} - \frac{1}{2} \right] + 1 \\
 &= \frac{3}{2}
 \end{aligned}$$

Q9

$$\begin{aligned}
 &\int_1^2 |2x - [3x]| dx \\
 &= \int_1^2 |\{3x\} - x| dx \\
 &= \int_1^2 (x - \{3x\}) dx \\
 &= \int_1^2 x dx - \int_1^2 \{3x\} dx \\
 &= \frac{(4-1)}{2} - 3 \int_0^{\frac{1}{3}} 3x dx \\
 &= \frac{3}{2} - \frac{1}{2} = 1
 \end{aligned}$$

Q10

$$\begin{aligned}
 1 &= \int_{-\pi}^{\pi} |\pi - |x|| dx \\
 &= 2 \int_0^{\pi} |\pi - x| dx \\
 &= 2 \int_0^{\pi} (\pi - x) dx \\
 &= 2 \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} \\
 &= 2 \left(\pi^2 - \frac{\pi^2}{2} \right) = \pi^2
 \end{aligned}$$

Q11

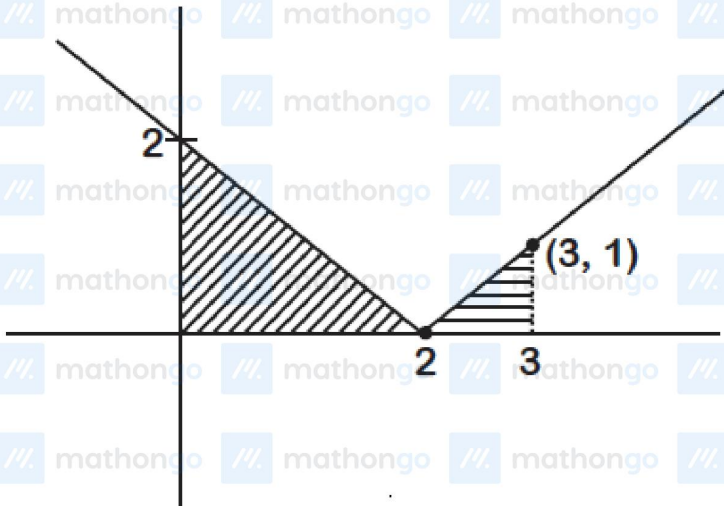
$$\begin{aligned}
 &\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx \quad \text{let } x = \sin \theta, dx = \cos \theta d\theta \\
 &= \int_0^{\pi/6} \frac{\sin^2 \theta \cdot \cos \theta d\theta}{\cos^3 \theta} = \int_0^{\pi/6} \tan^2 \theta d\theta \\
 &= \int_0^{\pi/6} (\sec^2 \theta - 1) d\theta = [\tan \theta - \theta]_0^{\pi/6} = \frac{1}{\sqrt{3}} - \frac{\pi}{6} \\
 \therefore \frac{k}{6} &= \frac{1}{\sqrt{3}} - \frac{\pi}{6} \\
 \Rightarrow k &= 2\sqrt{3} - \pi
 \end{aligned}$$

Q12

$$\because f(x) = |x - 2|$$

$$\text{So } \int_0^3 f(x) dx = \frac{1}{2} (2 \times 2 + 1 \times 1)$$

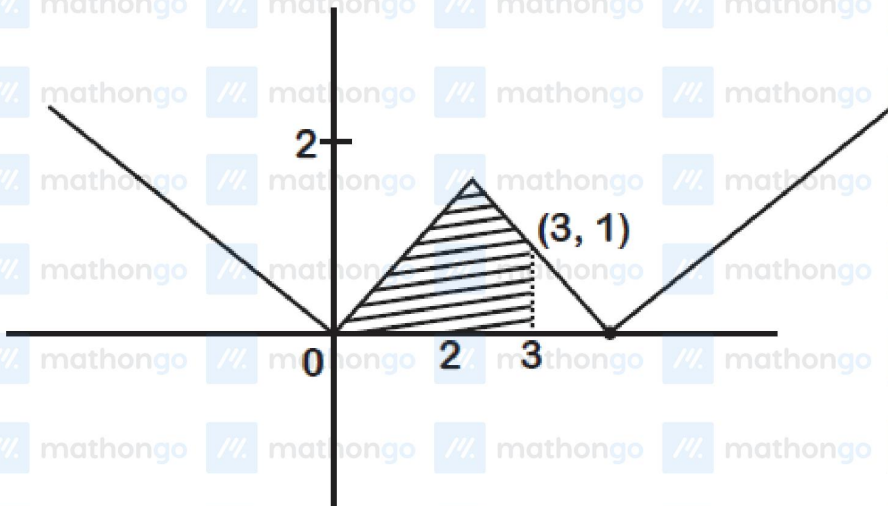
$$= \frac{5}{2}$$



$$\because g(x) = ||x - 2| - 2|$$

$$\text{So } \int_0^3 g(x) dx = \frac{1}{2} (2 \times 2) + \frac{1}{2} \times 1 \times (1 + 2)$$

$$= \frac{7}{2}$$



$$\text{Now } \int_0^3 (g(x) - f(x)) dx = \frac{7}{2} - \frac{5}{2} = 1$$

$$\begin{aligned} & \tan^3 x \sin^2 3x (2 \sec^2 x \sin^2 3x + 3 \tan x \sin 6x) \\ &= \frac{d}{dx} \left(\frac{\tan^4 x \sin^4 3x}{2} \right) \\ & \therefore \int_{\pi/6}^{\pi/3} \tan^3 x \sin^2 3x (2 \sec^2 x \sin^2 3x + \tan x \sin 6x) dx \\ &= \left. \frac{\tan^4 x \sin^4 3x}{2} \right|_{\pi/6}^{\pi/3} \\ &= \frac{9.0}{2} - \frac{\frac{1}{9} \cdot 1}{2} = \frac{-1}{18} \end{aligned}$$

Q14

$$\begin{aligned} \int_0^n \{x\} dx &= n \int_0^1 x \cdot dx = \frac{n}{2} \\ \int_0^n [x] dx &= \int_0^n (x - \{x\}) dx = \frac{n^2}{2} - \frac{n}{2} \\ \text{Given that } \frac{n}{2}, \frac{n^2-n}{2}, 10(n^2-n) &\text{ are in GP} \\ \therefore \left(\frac{n^2-n}{2} \right)^2 &= \frac{n}{2} \times 10(n^2-n) \\ \Rightarrow n^2 &= 21n \\ \therefore n &= 21 \end{aligned}$$

Q15

$$\begin{aligned} 1 &= \int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{\sin x}} dx \\ &= \int_0^{\pi/2} \left(\frac{1}{1 + e^{\sin x}} + \frac{1}{1 + e^{-\sin x}} \right) dx \\ &= \int_0^{\pi/2} \frac{1 + e^{\sin x}}{1 + e^{\sin x}} dx \\ &= \frac{\pi}{2} \end{aligned}$$

Q16

$$\begin{aligned} I_1 &= \int_0^1 (1 - x^{50})^{100} dx, I_2 = \int_0^1 (1 - x^{50})^{101} dx \\ \text{Here, } \alpha &= \frac{I_2}{I_1} \\ \text{Now, } I_2 &= \int_0^1 (1 - x^{50})^{101} dx = \int_0^1 (1 - x^{50}) (1 - x^{50})^{100} dx \\ I_2 &= \int_0^1 (1 - x^{50})^{100} - \int_0^1 x \cdot \frac{x^{49}(1-x^{50})^{100}}{\parallel} dx \end{aligned}$$

$$I_2 = I_1 + \left[\frac{x}{5050} (1-x)^{101} \right]_0^1 - \int_0^1 \frac{(1-x^{50})^{101}}{5050} dx$$

$$I_2 = I_1 + 0 - \frac{I_2}{5050}$$

$$\frac{5051}{5050} I_2 = I_1$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{5050}{5051} = \alpha$$

Q17

$$I = \int_1^2 e^x x^x (2 + \log_e x) dx$$

$$I = \int_1^2 e^x x^x [1 + (1 + \log_e x)] dx$$

$$= \int_1^2 e^x [x^x + x^x (1 + \log_e x)] dx$$

$$= [e^x x^x]_1^2, \left\{ \int e^x (f(x) + f'(x)) dx = e^x f(x) + c \right\}$$

$$= e^2 \times 4 - e \times 1$$

$$= 4e^2 - e$$

$$= e(4e - 1)$$